

MARWADI UNIVERSITY

DEPARTMENT OF AI, ML

DAA (01AI0506) - Lab Practical

Practical 05: Implementation of 0/1 Knapsack using Dynamic Programming

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Aim

Implementation of 0/1 Knapsack using Dynamic Programming as specified in the DAA Lab Manual.

Program

```
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;

int knapsack(int W, const vector<int>& wt, const vector<int>& val) {
    int n = wt.size();
    vector<vector<int>>> dp(n + 1, vector<int>(W + 1, 0));

    for (int i = 1; i <= n; i++) {
        for (int w = 0; w <= W; w++) {
            dp[i][w] = dp[i - 1][w];
            if (wt[i - 1] <= w)
                dp[i][w] = max(dp[i][w],
                               val[i - 1] + dp[i - 1][w - wt[i - 1]]);
        }
    }
    return dp[n][W];
}

int main() {
    vector<int> wt = {1, 3, 4, 5};
    vector<int> val = {1, 4, 5, 7};
    int W = 7;
    cout << "Maximum value: " << knapsack(W, wt, val);
}
```

Output

Maximum value: 9

Time and Space Analysis

Number of DP states is $(n+1)(W+1)$. Time Complexity: $\Theta(nW)$. Space Complexity: $\Theta(nW)$. The complexity is pseudo-polynomial because W is numeric capacity.

Result

The practical was implemented successfully and the expected output was obtained.